Phu Duong

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Education

Princeton University — Expected May 2028
Bachelor of Science in Engineering in Computer Science — *Princeton, NJ*
Minor in Statistics and Machine Learning

- Relevant Coursework: Data Structures and Algorithms, Advanced Algorithms, Distributed Systems, Operating Systems, Computer Architecture, Systems Programming, Machine Learning, Natural Language Processing, Computer Vision, Probability and Stochastic Systems, Statistics, Causal Inference
- Awards: QuestBridge National College Match Scholarship

Experience

Amazon Lab126 — June 2026 – August 2026
Software Development Engineer Intern — *Sunnyvale, CA*

- Tools: C++, Python
- Architecting the Smart Home Edge team's first centralized verification infrastructure for distributed, low-latency Alexa firmware across Echo, Fire TV, and eero, validating invariants, state consistency, and latency bounds

Princeton Computer Science — September 2025 – Present
Teaching Assistant — *Princeton, NJ*

- Tools: Java, C
- Led office hours for **175+ students** in COS 226: Algorithms and Data Structures, debugging Java implementations of union-find, kd-trees, dynamic programming, graph traversal, and MST-based clustering
- Led office hours for **120+ students** in COS 217: Introduction to Programming Systems, teaching low-level debugging, memory management, performance optimization, and systems security through C and ARMv8 assembly projects with GDB and Valgrind

NashTech — June 2025 – August 2025
Software Engineer Intern — *Ho Chi Minh City, VN*

- Tools: PHP, JavaScript, PostgreSQL, AWS
- Shipped an assignment management system for The Open University, unifying assignment creation, submission, and grading workflows previously scattered across **600+ course modules** used by **220,000+ students and educators**
- Optimized SQL queries for student progress data serving **60,000+ daily active users** by profiling bottlenecks with CloudWatch, applying database-side filtering, and adding a composite index, cutting execution time by ~**50%**

Jane Street — June 2024 – August 2024
Academy of Math and Programming Scholar — *New York, NY*

- Tools: Python
- Developed a high-frequency trading algorithm implementing statistical arbitrage and mean reversion strategies, generating **$5,000+ PnL** per round in Jane Street's Electronic Trading Competition
- Built decision-optimization algorithms for games, using Monte Carlo simulations to estimate expected values for Camel Up bets and minimax search to solve Wordle in ~**3.6 guesses**
- Ranked **1st place among 78 peers** in Jane Street's Estimathon and Mystery Planet math and trading competitions

Projects

Bolt-on Attention Module for Improved Mamba Recall | *PyTorch*

- Designed a lightweight **4.2M-parameter** cross-attention memory module, equivalent to only **0.06%** of a frozen Falcon Mamba 7B backbone, to improve long-context recall by storing hidden states from high-loss tokens in a bounded key-value cache
- Improved BABILong needle-in-a-haystack accuracy from **24% to 54%** across 0–16k context lengths, outperforming LoRA with 18× fewer trainable params while preserving O(n) inference and using 4,300× fewer FLOPs than a 128k-context Transformer

Natural Language Circuit Component Selector | *TypeScript, Python, React*

- Developed a multi-agent LLM system that converts natural-language hardware requirements into PCB component recommendations, reducing a component-selection bottleneck for electrical engineers from days to minutes
- Implemented a LangChain-based agent pipeline that converts hardware requirements into structured electrical specifications, retrieves compatible DigiKey components via a custom MCP server, and validates selections against dependency constraints
- **Won Best Business and Enterprise Hack** at HackPrinceton among **194 projects and 600+ participants**

Biometric-Controlled Melatonin Patch | *TypeScript, Python, C++, React Native*

- Designed a sleep-optimization platform that uses time-series classification on Fitbit biometrics to predict overnight wake risk and constrained optimization to generate melatonin-release curves under dosage, tapering, and hardware limits
- Engineered a Wi-Fi-enabled dispenser using Arduino and ESP8266, programming embedded C++ to process backend requests and control melatonin release
- **Achieved 2nd Best Hardware Hack** at HackPrinceton among **87 projects and 250+ participants**

Skills

Languages: Java, Python, C/C++, JavaScript/TypeScript, PHP, HTML/CSS, SQL
Frameworks & Technologies: PyTorch, Node.js, Express, Flask, React, React Native, AWS, Docker, Linux, Git